NOTRE DAME UNIVERSITY BANGLADESH

CSE LAB Report-02

Submitted by:

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Course Title: Data Structure Lab

Course Code: CSE-2104

Lab Task Topic: Array Insert, Delete & Sorting.

Submitted To:

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Problem No. 01

Name of the problem: Write a program in C to insert an element in an array.

Solution:

```
//Lab 2 task 1
// Insert an element
#include <stdio.h>
int main()
{
    int n, i, l, item;
    printf("Enter your array size:");
    scanf("%d", &n);
    printf("Enter Array elements:");
    int array[n];
    for (i = 1; i <= n; i++)
    {
        scanf("%d", &array[i]);
    }
    printf("Enter the element to be inserted in the array: ");
        scanf("%d", &item);
        printf("Enter the location where the element %d is to be inserted: ", item);
        scanf("%d", &l);
```

```
        if (l <= n)
        {
            int j;
            for (j = n; j >= l; j--)
            {
                array[j + 1] = array[j];
            }
            array[l] = item;
            n = n + 1;
        }
        else
            printf("The entered location is out of
bound.");
        printf("\nModified Array: ");
        for (i = 1; i <=n; i++)
            printf("%d ", array[i]);
        printf("\n");
}
```

Output:

```
"C:\Users\istia\OneDrive\Desktop\Cloud Computer\Code & Programs\C & C++\Array & Data Structure\Lab 2\20_05_Lab2_q1.exe"
Enter your array size:4
Enter Array elements: 12 23 34 45
Enter the element to be inserted in the array: 50
Enter the location where the element 50 is to be inserted: 3

Modified Array: 12 23 50 34 45

Process returned 0 (0x0)   execution time : 43.430 s
Press any key to continue.
```

Problem No. 02

Name of the problem: Write a program in C to delete an element in an array.

Solution:

```
//Lab 2 task 2
// delete element
#include <stdio.h>
int main()
{
    int n, i, l, item;
    printf("Enter your array size:");
    scanf("%d", &n);
    printf("Enter Array elements:");
    int array[n];
    for (i = 1; i <= n; i++)
    {
        scanf("%d", &array[i]);
    }
    printf("\nEnter the location from where element
is to be deleted: ");
        scanf("%d", &l);
        if (l <= n)
        {
            int j;
```

```
            item = array[l];
            printf("\nItem %d deleted from location
%d\n", item, l);
            for (j = l ; j < n ; j++)
            {
                array[j] = array[j + 1];
            }
            n = n - 1;
        }
        else
        {
            printf("\nThe Entered location is out
of bound.");
        }
        printf("Modified array: \n");
        for (i = 1; i <= n; i++)
        {
            printf("A[%d] = %d \n",i, array[i]);
        }
        printf("\n");
}
```

Output:

```
"C:\Users\istia\OneDrive\Desktop\Cloud Computer\Code & Programs\C & C++\Array & Data Structure\Lab 2\20_05_Lab2_q2.exe"
Enter your array size:4
Enter Array elements: 12 23 34 45

Enter the location from where element is to be deleted: 3

Item 34 deleted from location 3
Modified array:
A[1] = 12
A[2] = 23
A[3] = 45


Process returned 0 (0x0)   execution time : 16.913 s
Press any key to continue.
```

Problem No. 03

Name of the problem: Write a program in C to Sort a number of students by using their heights in an array.

Solution:

```
// Lab 2 task 3
// sort element
#include <stdio.h>
int main()
{
    int n, i, j;
    float t;

    printf("Number of Students : ");
    scanf("%d", &n);
    printf("Heights : ");
    float array[n];
    for (i = 0; i < n; i++)
    {
        scanf("%f", &array[i]);
    }

    for (i = 0 ; i < n - 1; i++)
    {
        for (j = 0 ; j < n - i - 1; j++)
```

```
        {
            if (array[j] > array[j+1])
            {
                /* Swapping */
                t         = array[j];
                array[j]   = array[j+1];
                array[j+1] = t;
            }
        }
    }
    printf("Sorted height: ");
    for (i = 0; i < n; i++)
    {
        printf("%.2f ", array[i]);
    }
}
```

Output:

```
"C:\Users\istia\OneDrive\Desktop\Cloud Computer\Code & Programs\C & C++\Array & Data Structure\Lab 2\20_05_Lab2_q3.exe"
Number of Students : 4
Heights : 5.3 6 5.8 5.5
Sorted height:
Students[0] = 5.30
Students[1] = 5.50
Students[2] = 5.80
Students[3] = 6.00
Process returned 0 (0x0)   execution time : 27.341 s
Press any key to continue.
```

Problem No. 04

Name of the problem: Write a program in C to Delete mth even element from an array.

Solution:

```
//Lab 2 task 4
// delete element of mth even value
#include<stdio.h>
int main(){
   int n, i, l, m, evenCount=0, item;
    printf("Enter value of m: ");
    scanf("%d",&m);
    printf("Enter your array size: ");
    scanf("%d", &n);
    printf("Enter Array elements: ");
    int array[n];
    for (i = 1; i <= n; i++){
        scanf("%d", &array[i]);
    }
    for (i = 1; i <= n; i++) {
        if (array[i] % 2 == 0) {
            evenCount++;
            if (evenCount == m){
               l =i;
               break;}
```

```
        }
    }
        if (l <= n){
            int j;
            item = array[l];
            for (j = l ; j < n ; j++){
                array[j] = array[j + 1];
            }
            n = n - 1;
            printf("The index of the mth even value
is %d\n", item);
        }
        if(l==0){
   printf("\nThere  is  no  even  value  in  this
array\n");}
        printf("Modified Array is: \n");
        for (i = 1; i <= n; i++){
            printf("A[%d] = %d \n", i, array[i]);
        }
        printf("\n");
}
```

Output:

```
"C:\Users\istia\OneDrive\Desktop\Cloud Computer\Code & Programs\C & C++\Array & Data Structure\Lab 2\20_05_Lab2_q4.exe"
Enter value of m: 2
Enter your array size: 4
Enter Array elements: 12 23 34 45
The index of the mth even value is 34
Modified Array is:
A[1] = 12
A[2] = 23
A[3] = 45


Process returned 0 (0x0)   execution time : 8.449 s
Press any key to continue.
```